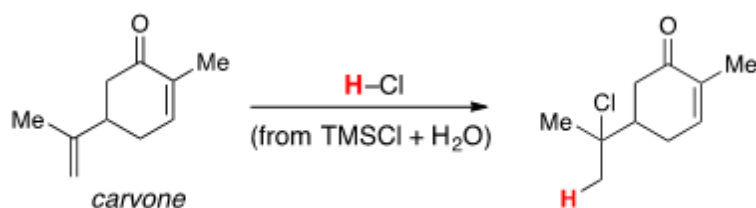


Activity 8: Alkene Addition Reactions: Selective Markovnikov Hydrohalogenation of Carvone

The Experiment



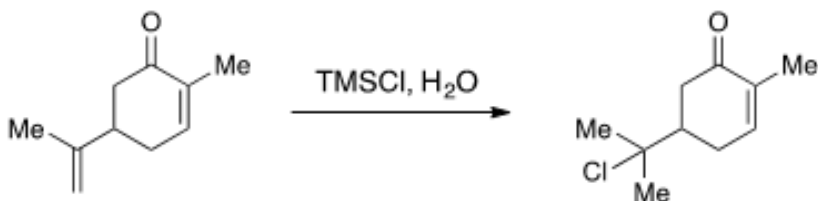
Lab Activity Goal

Verify experimentally the structural factors governing the chemo- and regioselectivity of alkene addition reactions in the context of the hydrochlorination of carvone.

Pre-Laboratory Assignments

See the Activity 8 Pre-Lab Preparation and Outline document in Labflow.

Experimental Procedure¹



Part 1. Carvone Hydrochlorination

1. Weigh a 50-mL round bottom flask.
 2. Using a clean plastic pipette, add 2 mL of carvone to the weighed 50-mL flask. Note that plastic pipettes have a 1 mL volume graduated into 0.25 mL increments.
 3. Weigh flask again to obtain the exact mass of carvone.
 4. Add 1 mL of H₂O to the carvone.
 5. Add a stir bar to the flask. Clamp the flask to the monkey bars on top of a stir plate.
 6. Obtain ~75 mL of saturated NaHCO_{3(aq)} in a beaker. Set a plastic pipette in the beaker and place in your hood for later use to quench/rinse glassware with acidic reagents.
- **Note:** Designate another beaker to serve as an aqueous waste that will later be transferred to the **aqueous** waste in the back of the laboratory.

Activity 8: Procedure

7. Obtain TMSCl (chlorotrimethylsilane) according to the directions below.



Important: Safety Pause

Note that **TMSCl is corrosive**. Carefully follow directions below and from your instructor. As always, wear goggles and gloves. If TMSCl comes in contact with your gloves or clothing, they must be immediately removed.

Safety Pause

- Set a **dry** 10-mL graduated cylinder in a 250-mL beaker. Invert a 50-mL beaker on top of the graduated cylinder to serve as a lid. Take the assembly to the TMSCl dispensing location.
 - Your instructor will add ~6 mL of TMSCl to your graduated cylinder. Holding the assembly with both gloved hands and being careful not to trip or bump into anyone, carefully take the TMSCl back to your hood.
8. Add the TMSCl to the stirring carvone/water mixture. Use a funnel to ensure that all the TMSCl is added to the inside of the flask. If a pipette is used to transfer the TMSCl, be sure to hold it vertically (not holding the pipette vertically can cause pipette contents to unexpectedly squirt out).
 9. **Record the time** that you finish adding the TMSCl.
 - **Note:** As always, make sure that you are wearing gloves for step 10. TMSCl reacts with water to make hydrochloric acid in the reaction flask.
 10. To clean up the residual TMSCl, pull the hood sash down as far as possible. Place the funnel on the graduated cylinder and rinse the funnel and the graduated cylinder with saturated $\text{NaHCO}_{3(\text{aq})}$ using the pipette. Place the (probably fizzing) grad cylinder in the back of your hood. Once fizzing stops, place the rinse from the graduated cylinder into your designated aqueous waste beaker.
 11. Be sure to dispose of your gloves properly. Get new gloves before you handle chemicals again.

Activity 8: Procedure

Part 2. Monitoring the Reaction Using TLC Analysis

Note: Use ~20 mL of the already prepared 30% EtOAc (ethyl acetate) in hexanes as the TLC developing solvent. A single TLC chamber can be used to run all of your TLC plates.

1. Prepare a UV-active TLC plate with 3 lanes.
 - Lane 1 should be the carvone standard starting material (S), lane 2 should be a co-spot (C), and lane 3 should be the reaction aliquot (R). The co-spot is made by spotting both the starting material and the reaction aliquot in the co-spot lane. **Be sure to use a clean capillary tube for the reaction aliquot.** Dispose of any used capillary tubes in the glass waste. If your capillary tube has liquid in it, gently tap out the excess liquid onto a paper towel before disposal.
 - For a reaction aliquot, use a pipette to obtain one drop of the reaction mixture. Add this drop to a small vial and dilute with two to four drops of TLC eluent. Use the diluted reaction aliquot for TLC analysis.

Pro-tip

Always use a clean capillary tube when spotting a new aliquot on a TLC plate. Even a small amount of contamination from a previous spot can make TLC results hard to analyze!

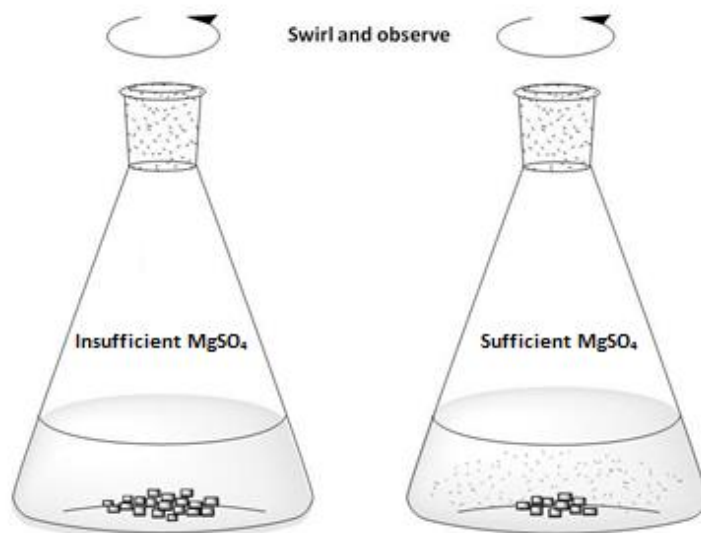
2. Do your first TLC analysis as soon as you have your plate prepared. Record the elapsed time of your reaction.
3. Monitor the progress of the reaction by doing a TLC analysis every 15 min. Repeat the TLC analysis until carvone is no longer observed in the reaction aliquot. The total reaction time is expected to be between 15-60 minutes. Record the time when the reaction is observed (by TLC) to be complete. Make sure you dispose of any used capillary tubes in the glass waste before you move on to Part 3.

Activity 8: Procedure

Part 3. Reaction Work-Up

1. Once TLC indicates that no carvone remains, add 5 mL of H₂O and 10 mL of diethyl ether to the reaction mixture.
2. Transfer the mixture to a clamped 250-mL Erlenmeyer flask. Via pipette carefully add ~5 mL NaHCO_{3(aq)}. Gently stir using stir bar and stir plate. If there is still observable fizzing, then carefully add more NaHCO_{3(aq)} in 5 mL portions until there is no fizzing, which should indicate the mixture is neutralized. Test the pH of the mixture with Congo red paper to ensure the mixture is no longer strongly acidic. A blue Congo Red test result means the mixture is still strongly acidic.
 - You can test the pH multiple times on a single strip of Congo Red paper. Set the Congo Red test strip on a paper towel, then lightly dip a stir rod into the mixture. Gently transfer a drop of the mixture on the filter paper. Repeat as often as necessary until the mixture is neutralized.
3. Transfer the quenched reaction mixture to the 125-mL separatory funnel. Let the layers separate. Drain the aqueous (denser) layer.
4. Wash the organic layer that remains in the separatory funnel with aqueous NaHCO₃ (10 mL). Once the layers separate and the aqueous layer is once again drained, wash the remaining organic layer with 10 mL of brine, i.e. saturated aqueous sodium chloride (10 mL). Let the layers separate and drain the brine.
5. Transfer the washed organic layer from above into an appropriately-sized Erlenmeyer flask and add a **small** scoop of MgSO₄ to dry. You will know the organic solution is dry when the some of the MgSO₄ flows freely upon swirling (see figure below).

Activity 8: Procedure



When the solution is dry (i.e. has no water remaining), not all of the MgSO_4 will settle at the bottom. Some MgSO_4 will stay suspended in solution and appear as a "snow globe".

6. Add 3 boiling stones to a 50-mL beaker. Weigh and record its mass.
7. Use gravity filtration to transfer the dried organic extract into the weighed beaker containing the boiling stones. Evaporate the ether from your carvone addition product using a steam bath.
 - Note that placing the steam bath on top of a lowered lab jack, rather than leaving it on the bench top, helps condensed water to drain into the cup sink (water runs downhill).
8. Once all the ether has evaporated, cool the beaker, and then record its mass.

Part 4. Product Characterization and Spectroscopic Analysis

1. **Finishing Activity 7:** obtain an IR spectrum, melting point range, and mass of your recrystallized Grignard reaction product.
2. Obtain an IR spectrum of **both** your reaction product and carvone.

Activity 8: Procedure

3. Your instructor will give you guidance on the analysis of the **carvone** starting material NMR spectrum so that you can complete the Activity 8 Post-Lab Assessment.

NOTE: The NMR spectrum of the carvone addition product (Spectrum A) is embedded in the Activity 8 Post-Lab Assessment. You will analyze this spectrum. Remember that methyl groups, like methylene hydrogens, can be diastereotopic even if they seem equivalent.

Before you leave lab, upload your in-lab notebook pages to Labflow. Your in-lab notebook pages must include the following in a **single pdf file**:

- Any experimental observations (including but not limited to: color changes, observational notes, measurements, sketches/images of your TLC plates, etc.)
- Your handwritten answers to the questions posted in the In-Lab Notebook Pages tile on Labflow
- A picture of your Organic Lab Cleaning Checklist

Clean-Up and Waste Disposal

Activity 8 Specific Clean-Up

- **Organic Liquid Waste:**
 - Acetone rinse from glassware washing (includes the final product)
 - TLC eluent
- **Aqueous Waste:**
 - NaHCO_3 solution waste (from quenching excess TMSCl **and** reaction mixture)
- **Solid Waste:**
 - MgSO_4
 - Used Congo Red paper
 - Used filter paper
 - Used gloves
- **Glass Waste:**
 - Acetone-rinsed vials used for TLC monitoring
 - Used capillary tubes

Routine Clean-Up

- Clean up your hood space of all trash, spilled chemicals, etc. Don't leave a mess for the people who share this hood at other time slots. Up to 24 other people are sharing this space at various time slots throughout the week. Clean up your mess!
- No dirty or clean glassware should be left in the sink. No used filter paper or wadded up dirty paper towels should be left in the sink.
- **Dirty paper towels go into the REGULAR TRASH**, not the red solid waste trash. This saves the university a lot of waste disposal money!
- Dispose of dirty filter paper in the red solid waste container.
 - Do **NOT** deposit filter paper in the liquid waste eco-funnel as this will cause a clog which will lead to a messy overflow of waste .
 - Do **NOT** leave filter paper in the sink as this is poor lab etiquette and rude.

Activity 8: Procedure

- Used gloves should be disposed of into the red solid waste.
- **Ask your instructor if you are in charge of cleaning up the balance area** for this week. If you are in charge do the following:
 - Be sure all reagent bottles are properly capped.
 - Use the small brush to clean off balances.
 - Use small broom and dustpan (located on the wall surrounding the balance area) to remove excess solid from counter. Dispose of excess solids in red solid waste container. Return broom and dustpan to proper location.
 - Wipe down the counter with a moist paper towel.
- **Ask your instructor if you are in charge of making sure the sink area is presentable** for the next lab group.
 - Wash any dirty glassware rudely left in the sink. Let clean glassware air dry on rack or window sill area.
 - Dispose of any paper towels, filter paper, or cotton rudely left in the sink.
 - **Do not remove broken glass from the drain or the sink.** Rather, tell your instructor and fill out a blue "something wrong form" located in the cabinet above the solid waste containers. Be sure to record room number and the Left/Right location of the sink. Give the form to your instructor who will pass it to the appropriate person.

Post-Lab Assignment

The Activity 8 Post-Lab Assessment is posted on Labflow and is due before the deadline posted on Labflow.

References

¹ Boudjouk, P.; Kim, B.-K.; Han, B.-H. *J. Chem. Educ.* 1997, 74, 1223-1224.